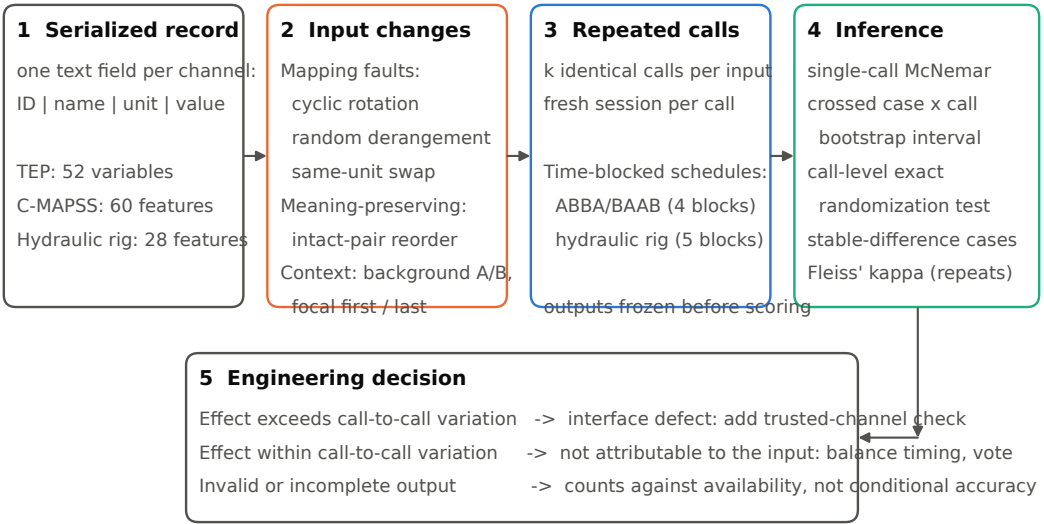
1 Serialized record
one text field per channel:
ID | name | unit | value
TEP: 52 variables
C-MAPSS: 60 features
Hydraulic rig: 28 features
2 Input changes
Mapping faults:
cyclic rotation
random derangement
same-unit swap
Meaning-preserving:
intact-pair reorder
Context: background A/B,
focal first / last
3 Repeated calls
k identical calls per input
fresh session per call
Time-blocked schedules:
ABBA/BAAB (4 blocks)
hydraulic rig (5 blocks)
outputs frozen before scoring
4 Inference
single-call McNemar
crossed case x call
bootstrap interval
call-level exact
randomization test
stable-difference cases
Fleiss' kappa (repeats)
5 Engineering decision
Effect exceeds call-to-call variation -> interface defect: add trusted-channel check
Effect within call-to-call variation -> not attributable to the input: balance timing, vote
Invalid or incomplete output -> counts against availability, not conditional accuracy